

FaceRoll Project Plan

Document Control

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Document History

Version	Issue Date	Changes
[1.0]	2/12/2026	Entire Document
[1.1]	8/27/2026	Updated Project plan to reflect the addition of geolocation and the removal of Canvas integration.

Document Approvals

Role	Name	Signature	Date
Project Sponsor <i>(faculty or external sponsor)</i>	Diana Rabah		8/27/2026
Project Review Group <i>(team members)</i>	Isaiah Bailey, Moses Ojo, Austin Woodruff, Cade Powell		2/12/2026
Project Manager	Gabriel Oniyitan		2/12/2026
Quality Manager <i>(if applicable)</i>			
Procurement Manager <i>(if applicable)</i>			
Communications Manager <i>(if applicable)</i>			
Project Office Manager <i>(if applicable)</i>			

1.1 Project Purpose, Goals and Objectives

Project Purpose	
The purpose of FaceRoll is to design and develop a prototype face-recognition-based attendance system that automates classroom attendance tracking in higher education environments. The system aims to improve attendance accuracy and reduce manual administrative effort for instructors, and provide a secure, privacy-conscious solution that operates independently from campus systems. FaceRoll will include several features that are designed to improve the attendance workflow. The system will enable instructors to manually change students' attendance per session. The system will use facial recognition to automatically mark and record students' attendance. This project also aims to have exportable attendance reports in CSV format and includes geolocation fencing, allowing the instructor to set a defined classroom location so students will have to be physically inside the classroom to be able to take attendance. The client requested adding this feature to address false check-ins. With this new feature, geolocation improves the system by providing another way to verify that students are actually within the classroom when checking in. This will help reduce false positives and remote check-ins and make the attendance record more accurate and reliable.	
Goal 1	Develop a functional prototype attendance system.
	Ensure facial recognition has good accuracy in multiple conditions
Goal 2	Ensure System reliability and privacy compliance.
Goal 3	Implement geolocation to accurately verify students are within the classroom zone.

1.2 Project Success Criteria

Category	Success Metric	Measurement Method	Target	Assigned to
Scope	All in-scope features implemented and demonstrated, including facial recognition, automatic attendance, manual overrides, instructor dashboard, student mobile app, attendance reporting, and geofencing, are all functioning as designed.	Comparison of completed system features against the approved SRS and WBS	100% of in-scope prototype requirements	Entire project team
Schedule	All sprint milestones, including facial recognition, geolocation	Review of Trello board deadlines against the syllabus schedule	All deliverables completed by the assigned due dates	Entire project team

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	development, testing, and integration, are completed according to the syllabus and Trello			
Budget	Project completed using free or academic-tier tools, including any libraries for facial recognition or geolocation, and utilizing GPS and camera-enabled devices.	Review of infrastructure, tools, and research findings to identify any additional changes to budget.	\$0 additional cost	Entire project team
Quality	FaceRoll operates without major functional failures across all core features such as facial recognition, automatic attendance, manual attendance override, student mobile app, instructor dashboard, attendance functions, geolocation fencing	Unit testing, feature testing, integration testing, geolocation validation, and live demo testing	All core systems perform successfully during the final presentation	Entire project team
Stakeholder satisfaction	Project Sponsor confirms all features within the system, including their requested feature, meet expected functionality	Feedback during sprint reviews and final demo	Sponsor confirms system meets project objectives	Entire project team
Risk management	Major risks like API limitations, facial recognition accuracy, geolocation accuracy, device permissions and compatibility, and privacy concerns have been well documented and addressed.	Risk log review	All high-priority risks are addressed before integration	Entire project team
Resource utilization	Workload is evenly distributed across all team members; geolocating fencing and facial recognition will not require any cost but will require GPS and camera-enabled devices.	Review of Trello board and sprint activity logs	Balanced task distribution across team members	Entire project team
Compliance	Facial data remains stored locally within project databases. The system shall only	Review test cases. Privacy settings and app permissions.	Location access must be requested from the user, and with their consent,	Entire project team

	obtain user location for attendance check-in purposes after the user grants permission to the application.		it should only be used for attendance check-in. No location data is stored. Facial data is not stored within the cloud and is only held on-prem/locally within computer storage.	
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1.3 Project Team

Name	Role	Responsibilities
Isaiah Bailey	Team Member	Instructor Dashboard, Facial recognition
Moses Ojo	Team Member	Assist with development and testing
Gabriel Oniyitan	Project Manager	Facial recognition/Instructor Dashboard
Cade Powell	Team Member	Mobile/Student app, facial recognition
Austin Woodruff	Team Member	Assist testing, integration, and system support. Mobile/Student App Assist

1.4 Project Governance (use N/A where not applicable to you)

Area	Description	Responsible Party/Role
Steering Committee	N/A	
Change Control Board	Changes to the scope will be discussed and approved by the team and the project sponsor.	Entire Project Team
Compliance Auditing	N/A	
Stakeholder Communication	Weekly team meetings, Discord, communication, and instructor feedback during sprint reviews	Entire Project Team
Performance Reviews	Sprint-based deliverable reviews, peer evaluations, and instructor grading	Entire Project Team
Project Closure Review	Final Presentation, Prototype Presentation	Entire Project Team

1.5 Stakeholder Management Plan

Name	Role	Power	Interest	Contact information
Diana Rabah	Instructor/Sponsor	High	High	Diana.Rabah@unt.edu
Project Developers	Developers	Medium	High	Discord group channel
Instructors	System users	Medium	High	N/A
Students	System Users	Low	High	N/A

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Level of Influence	High			
	Medium			Project Sponsor
	Low			Project Developers / Future Instructors
				Students
		Low	Medium	High

Level of Interest

1.6 Scope Management Plan

Scope Baseline			
Task	Dependencies	Deliverables	Acceptance Criteria
Finalize System Requirements and architecture	N/A	Updated requirements, design documents, and SRS	Requirements were agreed upon by the team and project sponsor
Develop and integrate facial-recognition logic	Project requirements and system design - Finish-to-start	Face detection, face matching logic, confidence threshold	Students can enroll their facial data, and the system can identify enrolled students
Develop attendance system and instructor dashboard	Facial recognition and database design - finish-to-start	Attendance sessions, present/late/absent statuses, instructor dashboard, manual attendance override, attendance storage and reports/	The system will be able to automatically record attendance for recognized students, and instructors will be able to view attendance by session and modify attendance as well.
Develop student mobile app functionality.	System design, authentication, facial recognition - finish-to-start	Functional mobile app for student authentication, facial enrollment, and attendance history	Students can authenticate, enroll their face, and interact with the attendance system.
Implement classroom geofencing	Student mobile app, attendance-session functionality - Finish-to-Start	Location-based attendance	The instructor can define an allowed classroom area where the system will allow the student to check in depending on whether they are in the classroom.

Testing and Validation	Core system components implemented - Finish-to-start	Requirement-based cases, usability performance testing	test and	Requirements pass their defined test; any major defects will be resolved.
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Exclusions, Assumptions and Constraints	
Exclusions	Integration with official UNT authentication Systems Access to campus ID photo databases Biometric compliance certifications Canvas integration has been removed from project scope due to university privacy reasons.
Assumptions	Students have GPS and camera-enabled devices. Firebase free tier is sufficient Classroom internet connectivity is available Instructors will have access to a camera of adequate quality The team can use an existing face recognition model/library Student rosters and course sections are maintained in Canvas and are accurate enough to sync (rejected feature) The school allows the use of face recognition software
Constraints	Limited API access No institutional admin privileges Budget limited to \$0 Face recognition accuracy may be affected by lighting, camera placement, and student movement. Geolocation accuracy could vary between devices, indoor conditions, and OS permissions. Semester timeline

Milestone	Description	Delivery Date
Sprint 1	Gathering Initial Project Requirements, Feasibility study of features, creating SRS, Project Charter, Project plan, WBS, and User Stories	02/22/2026
Sprint 2	Design phase of project, creating Design Document 1 & 2.	03/22/2026
Sprint 3	Testing plan; continue development of project. Presented the project to the sponsor	04/12/2026
Sprint 4	Continuing Development of FaceRoll by adding a new feature: Geolocation fencing to ensure students are within classroom boundaries, preventing abuse of attendance scoring.	9/06/2026

1.7 Resource Requirements

Task	Resource requirements
Facial recognition research	Internet research resources
Geolocation Research	Mobile devices with GPS services, React native location-permission library, classroom location data, Firebase integration.
Create Software Requirements Specification (SRS)	Assigned student author, SRS template, collaboration tools, version control (GitHub)

Test case creation	Requirements document, testing templates, team review sessionsRequirements document, testing templates, team review sessions
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1.8 Project Schedule

The FaceRoll project will be completed across multiple development sprints throughout the semester. The schedule aligns with the major deliverables outlined in the WBS and course timeline.

Semester 1:

Phase 1- Requirements and Planning

Duration: Weeks 1-3

Milestones:

- Requirement Gathering
- SRS Document
- WBS Document
- Trello Board

Phase 2 - System Design

Duration: Weeks 3-5

Milestones:

- Database architecture finalized
- API architecture defined
- Face recognition library selected
- UI/UX wireframes completed

Phase 3 - Core Development

Duration: Weeks 5 - 9

Milestones:

- Student Authentication implemented
- Face enrollment and capture functionality
- Face matching logic implemented
- Instructor Dashboard initial version complete

Phase 4 - Integration and Enhancement

Duration: Weeks 9 - 12

Milestones:

- Grace period & Late detection implemented
- Opt-Out & Privacy Controls Functional
- The attendance export feature added

Phase 5 - Testing & Validation

Duration: Weeks 12 - 14

Milestones:

- Unit testing completed
- Integration testing completed
- Non-functional validation completed

Phase 6 - Final Demo

Duration: Final Week

Milestones:

- Final Prototype Presentation

Semester 2:

Phase 1: New Features Proposal and Continuing Development

Milestones:

- Complete Change Request Form for requested feature.
- Canvas integration has been removed from project scope following sponsor feedback.
- Feasibility Study for Geolocation feature
- Update documentation to reflect new features.

Phase 2: Development and Integration

Milestones:

- Refine and add any features that weren't completed from last semester
- Implement Geolocation fencing
- Integrate within student check-in
- Add location permissions

Phase 3: System Testing and Refining

- Test geolocation accuracy
- Test facial recognition and attendance workflow
- Fix bugs and improve reliability

Phase 4: Final Integration and Documentation

- Complete final documentation
- Finalize the System
- Prepare for final demonstration

Phase 5: Final Testing and Presentation

- Complete final system testing
- Conduct final prototype demo

1.9 Project Budget

Task	Labor Costs	Material Costs	Other Costs	Budget	Actual
All Task	\$0	\$0	\$0	\$0	\$0
Geolocation Fencing	\$0	\$0	\$0	\$0	\$0

1.10 Change Management Plan

Change Control Board

Name	Role	Responsibilities
Isaiah Bailey	Project Developer	Reviews and approves changes, implements approved updates, ensures technical feasibility of requested changes
Moses Ojo	Project Developer	Reviews change requests, assists in impact analysis, supports implementation and testing of approved changes
Gabriel Oniyitan	Project Manager	Oversees change control process, coordinates approvals, ensures changes align with project scope and schedule
Cade Powell	Project Developer	Evaluates impact on mobile app and geolocation features, assists with implementation and testing
Austin Woodruff	Project Developer	Supports system testing, validates changes, ensures updates do not break existing functionality

Change Log

Date Identified	Request	Impact	Approval	Date Started	Date Completed
8/20/2026	Add geolocation fencing as a feature to the attendance workflow	Moderate	Approved by the entire project team and requested by the sponsor	8/20/2026	N/A
8/20/2026	Canvas Integration has been removed by the project team due to feedback from the sponsor due to university privacy concerns.	Moderate	Approved by the entire project team and requested by the sponsor	8/20/2026	8/20/2026

1.11 Risk Management Plan

Risk Matrix	Severity				
	Insignificant	Minor	Moderate	Major	Severe

Likelihood	Almost Certain	Medium	High	Very High	Very High	Very High
	Likely	Medium	High	High	Very High	Very High
	Possible	Low	Medium	High	High	Very High
	Unlikely	Low	Low	Medium	Medium	High
	Rare	Low	Low	Low	Low	Medium

Risk	Likelihood	Impact	Response	Risk Level	Risk Owner
Face recognition accuracy issues	Likely	High	Allow manual instructor override	High	Isaiah Bailey
Geolocation accuracy	Possible	High	Hard-code GPS coordinates for classrooms	Medium	Cade Powell

1.12 Quality Management Plan

Task or Deliverable	Quality Standards	Quality Assurance Guidelines	Quality Control Procedures
Adding geolocation fencing to classroom attendance	Feature must be able to determine whether a student is within the instructor-defined boundary of the classroom before allowing the student to check in.	Testing under different classroom locations, devices, and permission settings	Verify that students inside the geofence can check in and students that are outside are unable to check in.
Location access permissions	Location access must only be used with the user's consent and when needed for attendance check-in.	Review app permission handling to make sure location is requested only when attendance check-in happens	Test permission approval and denial to confirm the app behaves correctly
Facial Recognition	The system should correctly identify students with acceptable accuracy	Test recognition within different conditions like different lighting and camera positions.	Compare results with enrolled students and document any errors to be fixed.

Manual attendance	The instructor must be able to manually change the attendance of a student when needed.	Test attendance statuses and confirm changes are saved throughout the whole system.	Verify that updates appear correctly after editing.
Student Mobile App	Students should be able to authenticate, enroll their face, and view attendance information.	Test major functions on mobile devices	Verify all features of the student mobile app are working as designed.
Attendance reports	Exported attendance reports should obtain accurate and readable data	Compare exported reports with records that are stored within the system.	Open CSV export files and verify all contents are correct

1.13 Appendix

Appendix A. Feasibility study for approved feature.

Cost: Low

Expo is free and has a location library that can be used with Expo Go, so there should be no need for any third-party location service, meaning this feature should be free.

Time Impact: Medium

Since Expo has a location library as mentioned, implementation shouldn't take too long.

The main features would be:

- Establishing the location of the classroom and a permitted geofence radius
- Calculate student distance and ensure accurate location
- Manual testing- this is the part that will take the longest since we will have to perform regular checks any time new features are added

Benefit:

Geofencing would improve the attendance verification of FaceRoll. Since we already have facial recognition that verifies the student is who they say, the geofencing would ensure that said student is actually in class.

Appendix B: Geolocation Change Request Form**CHANGE REQUEST FORM****Change Description**

Project Name: FaceRoll - Face Recognition Attendance System	Change Name: Geolocation Fencing for Classroom Attendance	Number: 1
Requested By: Instructor	Contact: Capstone Project Team	Date: August 23, 2026
Description of Change: Incorporate geolocation fencing so that attendance is only recorded when a student is inside the designated geofence or classroom.		
Reason for Change: To ensure that students are physically present in the classroom when attendance is recorded, the instructor would like an extra location check.		
Priority [Circle One]: 1. High 2. Medium 3. Low		
Impact on Deliverables: includes new testing for classroom borders, user authorization handling, location validation rules, and a geolocation requirement.		
Impact of Not Responding to Change (and Reason Why): Without this modification, FaceRoll would only use facial recognition to confirm that attendance activities take place in the classroom.		
Date Needed:	Approval of Request: Project Team / Instructor	Date: August 23, 2026

Change Impact

Tasks/Scope Affected: Attendance workflow, system requirements, privacy requirements, UI messages, and test cases.
Cost Evaluation: Low financial impact; additional team development and testing time is required.
Risk Evaluation: Medium. Location accuracy, denied permissions, and GPS limitations indoors could affect results.
Quality Evaluation: Can improve attendance verification, but the feature must be tested to avoid incorrectly blocking valid attendance.
Additional Resources: Location-capable test devices, geolocation APIs/services, test classroom locations, and team testing time.
Duration: Estimated 1 sprint for implementation and initial testing.
Additional Effort: Moderate additional design, implementation, and testing effort.
Impact on Deadline: May add work to the Capstone II schedule but is manageable if prioritized early.
Alternative and Recommendations: Use a reasonable geofence radius and provide a clear error message when location cannot be verified.
Comments: Accept the change and add it to the Capstone II backlog.

Sign Offs

[Circle One]: 1. Accepted 2. Deferred 3. Rejected 4. More Info Requested
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Appendix C: Canvas Integration Change Request Form

CHANGE REQUEST FORM

Change Description		
Project Name: FaceRoll - Face Recognition Attendance System	Change Name: Canvas LMS Integration	Number: 2
Requested By: Instructor	Contact: Capstone Project Team	Date: August 23, 2026
Description of Change: To enable FaceRoll to transmit or synchronize student attendance records with Canvas, add Canvas LMS connectivity.		
Reason for Change: In order to facilitate the transfer and management of attendance records between FaceRoll and Canvas, the instructor desired Canvas integration.		
Priority: Medium		
Impact on Deliverables: If accepted, the project would need to synchronize attendance, integrate the Canvas API, and authenticate users. Nevertheless, the team lacks the necessary Canvas permissions and API access to put these capabilities into practice.		
Impact of Not Responding to Change (and Reason Why): Since attendance can still be tracked and managed within FaceRoll, there is no significant impact on the core FaceRoll technology. The only drawback is that instructors will have to handle attendance independently of Canvas.		
Date Needed: During Capstone II	Approval of Request: Project Team / Instructor	Date: August 23, 2026

Change Impact
Tasks/Scope Affected: System architecture, external interfaces, security/privacy requirements, attendance data handling, and testing scope.
Cost Evaluation: There is no out-of-pocket expense, but if Canvas access were granted, it would take more time to create and test the integration.
Risk Evaluation: The primary risk is that the integration cannot be implemented because the team lacks the necessary Canvas permissions or API access. Because FaceRoll would have to access and transfer student attendance and course information between systems, privacy and university data access issues might also arise.
Quality Evaluation: Could improve convenience, but external dependency and privacy concerns may reduce control over student data.
Additional Resources: Canvas API access, credentials, integration documentation, security review, and additional testing time.
Duration: Since implementation depends on obtaining the necessary Canvas permissions and API access, it is impossible to say.
Additional Effort: High additional effort because of authentication, API handling, privacy review, and failure handling.
Impact on Deadline: Could delay higher-priority FaceRoll features and increase project scope.
Alternative and Recommendations: Reject the Canvas integration for the current project. Keep attendance processing and records within FaceRoll to maintain the project privacy-focused scope.
Comments: The request is documented but will not be implemented because of privacy concerns and the added external-data dependency.

Sign Offs	
Decision: Rejected	
Comments: Because the team does not presently have the necessary authorization or API access from Canvas, the Canvas integration request was denied. Connecting FaceRoll to Canvas and managing student data may also raise privacy and university data-access issues. If the appropriate permits and approvals are given, the integration may be reexamined in the future.	
Project Manager Signature: Gabriel Oniyitan	Date: 8/23/26
Decision Maker Signature:	Date:

